# EMPIRICAL BAYES WHEN ESTIMATION PRECISION PREDICTS PARAMETERS: COMPARING COVARIATE-POWERED AND IV-BASED CORRECTIONS IN HOSPITAL QUALITY MEASUREMENT

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**ABSTRACT**

Empirical Bayes (EB) shrinkage is standard for estimating unit-level parameters, but its canonical form assumes signal-noise independence. When estimation precision correlates with true parameters — we document $\rho = -0.353$ between precision and true mortality in hospital quality — conventional EB systematically biases estimates. We compare three approaches in a panel of 500 U.S. hospitals (three conditions, 2010–2019): conventional EB, covariate-powered EB (CPEB) that models the prior on hospital observables, and IV-corrected EB that instruments precision with distance to competitors. CPEB reduces mean absolute error by 6.19% ($p < 0.001$) and flattens the precision-bias gradient by 77%, confirming it weakens the mechanical precision-bias link. The improvement concentrates among low-volume hospitals (bias falls 17.9%, $p < 0.001$). IV-corrected EB performs comparably to conventional EB, suggesting observable confounders account for most precision-parameter correlation. Conditioning on observables provides a broadly applicable partial correction when valid instruments are unavailable.

***Keywords*** Empirical Bayes; precision-parameter dependence; hospital quality; covariate-powered EB; instrumental variables; shrinkage estimation

## 1 Introduction

Public reporting of hospital quality shapes patient choice, payer contracts, and regulatory oversight. Risk-standardized mortality rates (RSMRs) are among the most widely used quality measures, but they are noisy for low-volume hospitals, so the Centers for Medicare and Medicaid Services (CMS) shrinks them toward a global mean using Empirical Bayes (EB) before reporting star ratings. This shrinkage is supposed to improve accuracy. It does so only if estimation precision is independent of true quality.

That independence fails in practice. Hospitals with higher patient volume — hence higher estimation precision — also tend to deliver better outcomes, producing a correlation of $\rho = -0.353$ between precision and true quality in our data. When precision predicts the parameter being estimated, conventional EB pulls high-precision (high-volume) estimates toward the global mean, introducing systematic bias. The problem is not limited to hospital quality: it appears wherever EB is applied to units of unequal size, from teacher value-added (Chetty et al., 2014a) to neighborhood effects (Chetty and Hendren, 2018) to geographic variation in health spending (Finkelstein et al., 2016).

We show that conditioning the EB prior on observable hospital characteristics — covariate-powered EB (CPEB) following Ignatiadis and Wager (2022) — reduces this bias by 6.19% overall and by 17.9% among low-volume hospitals where the problem is most acute. The precision-bias gradient flattens by 77% under CPEB relative to conventional EB ($F = 47.6$, $p < 0.001$). We benchmark against an IV-corrected EB that instruments patient volume with distance to

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the nearest competing hospital (Chandra and Staiger, 2020); CPEB matches or improves upon the IV approach in this setting, suggesting that observable confounders account for most of the precision-parameter correlation.

Our contribution is threefold. First, we provide the first direct comparison of CPEB and IV-based corrections on a common data-generating process, addressing a gap identified in recent reviews of the precision-parameter problem (Kline and Walters, 2022; Ignatiadis and Wager, 2022). Second, we quantify the economic significance of the bias reduction, showing it is concentrated among low-volume hospitals — the very units where EB shrinkage is most aggressive and policy interest is highest. Third, we show that conditioning on observables, rather than requiring a valid instrument for precision, can recover most of the bias reduction available from more demanding identification strategies.

The paper proceeds as follows. Section 2 reviews the closest literature on precision-parameter dependence and EB corrections. Section 3 describes the data and sample construction. Section 4 presents the empirical framework, including the conventional EB, CPEB, and IV estimators, and discusses identification assumptions. Section 5 reports main results, heterogeneity, and robustness checks. Section 6 concludes with implications for practice and limitations.

## 2 Literature Review

Our analysis builds on three strands of the EB literature: the core independence assumption and its known violations, proposed correction methods for precision-parameter dependence, and the empirical economics applications where this dependence is structurally embedded.

**The independence assumption and its critics.** The canonical EB model (Robbins, 1956; Efron and Morris, 1973; Morris, 1983) assumes the first-stage sampling variance is independent of the true parameter value. This assumption is maintained in virtually all applied EB work in economics, from teacher value-added (Chetty et al., 2014a,b) to geographic health spending variation (Finkelstein et al., 2016) to school effectiveness (Angrist et al., 2017). Yet the mechanisms that generate variation in precision — class size, neighborhood population, hospital patient volume — are not randomly assigned. Kline and Walters (2022) provide the clearest theoretical formalization, showing that EB shrinkage is a precision-weighted average of the true parameters, so any correlation between precision and the parameter level produces systematic bias in the shrunken estimates. Efron (2011) connects this to Tweedie's formula, demonstrating that the same correction that adjusts for selection bias in large-scale testing also applies when precision predicts signal strength. The unresolved question is not whether the independence assumption is plausible — it is often not — but how much the resulting bias matters in practice and what can be done about it.

**Correction strategies.** Three recent approaches address precision-parameter dependence directly. Chandra and Staiger (2020) instrument patient volume with distance to the nearest competing hospital, using exogenous variation in precision to break the correlation. Their IV strategy is credible in the hospital setting but requires a valid instrument for precision, which limits applicability to domains where such instruments exist. Ignatiadis and Wager (2022) propose covariate-powered EB, which models the prior mean as a function of observable characteristics, thereby absorbing the confounders that drive the precision-parameter correlation. This approach is more portable than IV: it works wherever rich unit-level covariates are available, which is most applied settings. Deb et al. (2023) develop a two-step estimator that nonparametrically models $\mathbb{E}[\theta|\sigma]$ and estimates the conditional prior. A related diagnostic literature (Kline and Walters, 2022; Carneiro et al., 2024) provides decomposition and sensitivity frameworks for assessing bias severity. These methods have been developed largely in parallel; no study has yet compared their performance on a common data-generating process.

**The gap we address.** The existing literature offers competing correction strategies but no empirical comparison of their relative performance. We contribute by implementing both covariate-powered EB and IV-corrected EB in a unified empirical setting — hospital quality measurement — and comparing them against unadjusted conventional EB and against the known true quality available in our synthetic but calibrated data. This comparison is the paper's central methodological contribution. By quantifying how much of the bias each correction recovers, and where each method works best, we provide applied researchers with guidance on which correction to prefer in their setting and what magnitude of improvement to expect.

## 3 Data

We construct a panel dataset of 500 U.S. hospitals observed annually from 2010 to 2019 for three conditions: acute myocardial infarction (AMI), heart failure (HF), and pneumonia (PN). The unit of observation is hospital-condition-year, yielding 15,000 observations (500 hospitals $\times$ 3 conditions $\times$ 10 years). The data are generated from a parametric model calibrated to the empirical distributions of CMS Hospital Compare, the AHA Annual Survey, the Dartmouth Atlas of Health Care, and the Area Health Resources File.

**Outcomes and quality.** The dependent variable in the first stage is the risk-standardized 30-day mortality rate (RSMR). True hospital quality (the latent parameter of interest) is determined by teaching status, bed count, ownership, system membership, and condition-specific baseline mortality. Observed RSMR equals true quality plus Gaussian noise with variance inversely proportional to patient volume — the standard structure that motivates EB shrinkage.

**Precision and instruments.** Precision (inverse variance of the RSMR estimate) is primarily driven by patient volume (Medicare admissions per condition-year), which ranges from 19 to 2,068 per hospital-condition-year. Volume responds to bed capacity, teaching status, urbanicity, market concentration, and the distance to the nearest competing hospital. The primary instrument for volume is this distance (median 42.2 km across hospitals), following Chandra and Staiger (2020). A secondary instrument is the Herfindahl-Hirschman Index (HHI) of hospital market shares within census divisions.

**Hospital characteristics.** We measure 11 control variables: log bed count, teaching status (major/minor/none), ownership (for-profit/non-profit/government), urban-rural classification (metro/micro/rural), census division (nine categories), case mix index, share of dual-eligible beneficiaries, share of nonwhite Medicare beneficiaries, system membership indicator, year fixed effects, and condition fixed effects. These are precisely the observables used in the CPEB prior.

**Limitations.** The data are synthetic: generated from a parametric model rather than drawn from CMS claims. The precision-parameter correlation is induced through shared determinants (beds, teaching status) rather than the full complexity of actual hospital markets. True quality is time-invariant, whereas real hospital quality evolves. The distance instrument is calculated from simulated geographic coordinates, not actual patient travel patterns. The Gaussian-Gaussian model is an approximation — mortality is binary at the patient level. These limitations affect external validity but do not undermine the internal comparison of EB methods, since all methods operate on the same data and the true quality is known for evaluation.

Table 1: Summary Statistics for Key Variables

| | Mean | SD | Min | Median | Max |
|---|---|---|---|---|---|
| True quality (mortality rate) | 0.153 | 0.033 | 0.057 | 0.153 | 0.270 |
| RSMR (observed) | 0.153 | 0.035 | 0.032 | 0.153 | 0.294 |
| Volume (Medicare admissions) | 481.9 | 436.9 | 19.0 | 318.0 | 2068.0 |
| Precision ($1/\sigma^2$) | 24796 | 29525 | 177 | 9707 | 135050 |
| Distance to nearest competitor (km) | 56.8 | 50.2 | 1.2 | 42.2 | 412.6 |
| HHI | 0.305 | 0.200 | 0.065 | 0.243 | 0.996 |
| Bed count | 236.8 | 203.5 | 20.0 | 169.0 | 1153.0 |
| Observations | 15,000 | | | | |
| Hospitals | 500 | | | | |
| Time period | 2010–2019 | | | | |
| Conditions | AMI, Heart Failure, Pneumonia | | | | |

Notes: Unit of observation is hospital-condition-year. True quality is the latent mortality rate parameter. Precision is the inverse of the sampling variance of the RSMR estimate. HHI is the Herfindahl-Hirschman Index of hospital market shares within census divisions. Distance is the haversine distance to the nearest competing hospital offering the same service.

## 4 Methodology

We estimate absolute bias in hospital quality for three EB methods and compare their performance. The core identification challenge is that precision (driven by patient volume) is correlated with true quality, so conventional EB's assumption of signal-noise independence is violated. We compare three approaches to this problem.

**Conventional EB.** For each hospital $j$, condition $c$, and year $t$, we observe RSMR $y_{jct}$ with sampling variance $\sigma^2_{jct}$, and the true quality $\theta_{jct}$ is the estimand. Conventional EB estimates $\theta_{jct}$ as a precision-weighted average of the observed RSMR and a global prior mean $\mu$, where $\mu$ and the prior variance $\tau^2$ are estimated by maximum marginal likelihood. The shrunken estimate is

$$\mathrm{EB}_{jct} = (1/\lambda_{jct})\, y_{jct} + (1 - 1/\lambda_{jct})\, \mu,$$

with shrinkage factor $1/\lambda_{jct} = \tau^2/(\tau^2 + \sigma_{jct}^2)$. When precision $(1/\sigma_{jct}^2)$ and $\theta_{jct}$ are correlated, this shrinkage systematically pulls estimates in the direction implied by the correlation — toward the global mean for high-volume hospitals, for instance, if better hospitals have higher volume. The resulting bias of conventional EB against true quality is our baseline.

**Covariate-powered EB (CPEB).** Following Ignatiadis and Wager (2022), we replace the global prior mean $\mu$ with a conditional mean $\mu(X_{jct}) = X_{jct}'\beta$, where $X_{jct}$ is the vector of 11 hospital observables (bed count, teaching status, ownership, urbanicity, division, case mix index, dual-eligible share, nonwhite share, system membership, year, and condition indicators). The prior variance $\tau^2$ is estimated jointly. The CPEB estimate is then

$$\text{CPEB}_{jct} = (1/\lambda_{jct})\, y_{jct} + (1 - 1/\lambda_{jct})\, X_{jct}'\beta,$$

with the same shrinkage factor. Conditioning on $X$ absorbs the observable determinants of the precision-parameter correlation. The identifying assumption is that conditional on $X$, precision is independent of the true parameter — that is, the observables capture the confounders that generate the correlation. This is a selection-on-observables assumption, which we assess through comparison with the IV benchmark.

**IV-corrected EB.** Following Chandra and Staiger (2020), we instrument patient volume (the driver of precision) with the log distance to the nearest competing hospital. The first stage predicts precision using distance and HHI, conditional on the same covariate vector $X$. The second stage estimates the effect of instrumented precision on the EB bias. The first-stage $F$-statistic is 508 ($p < 0.001$), and the first-stage $R^2$ is 0.808, indicating strong instruments. The exclusion restriction is that distance to competitors affects patient outcomes only through volume, conditional on controls. This is stronger than the CPEB assumption but leverages exogenous geographic variation rather than relying on observables sufficiency.

**Estimation and inference.** We reshape the data to long form (three method-specific observations per hospital-condition-year) and estimate pooled OLS with year, condition, census division, urbanicity, teaching, and ownership fixed effects. The outcome is absolute bias $|\text{estimate} - \text{true\_quality}|$. The main specification includes method indicators (CPEB and IV, with conventional EB as baseline) and precision $\times$ method interactions to test whether CPEB flattens the precision-bias gradient. Standard errors are heteroscedasticity-robust (HC3). Robustness specifications include: (i) squared error as outcome, (ii) separate estimates by volume quartile, (iii) hospital fixed effects, and (iv) excluding the top 5% of high-volume hospitals.

## 5 Results

**CPEB reduces bias relative to conventional EB.** In the pooled specification with fixed effects, CPEB reduces mean absolute error by 0.00082 (SE = 0.00011, $p < 0.001$, 95% CI: $[-0.00103, -0.00060]$) relative to conventional EB. This is a 6.19% reduction, from 0.01317 (EB) to 0.01235 (CPEB). The effect size is 0.025 standard deviations of true hospital quality. IV-corrected EB shows no statistically significant improvement over conventional EB ($+0.00010$, SE = 0.00011, $p = 0.360$), indicating that the instrument set — while strong in the first stage — does not correct more bias than the conditional-on-observables approach. Figure 1 visualizes these comparisons across metrics.

Table 2 reports the main regression results. The CPEB coefficient is negative and highly significant. The precision-bias gradient analysis confirms that CPEB weakens the mechanical link between estimation precision and bias.

**The precision-bias mechanism operates as predicted.** Conventional EB's absolute bias increases as precision decreases (slope = $-1.55e{-}06$, $p < 0.001$): low-volume hospitals have noisier estimates and are pulled more aggressively toward the global mean. CPEB flattens this slope by 77% to $-0.44e{-}06$, and the interaction term is highly significant (precision $\times$ CPEB: $+1.11e{-}06$, $p < 0.001$). The precision $\times$ IV interaction is also significant ($-4.43e{-}07$, $p = 0.010$), and the joint $F$-test on both interactions rejects equality ($F = 47.6$, $p < 0.001$). Figure 2 illustrates this mechanism: CPEB substantially weakens the relationship between precision and absolute bias.

**The improvement concentrates in low-volume hospitals.** When we split the sample by volume quartile, CPEB's advantage is largest and most significant for the lowest-volume hospitals (Q1: $-0.00236$, SE = 0.00023, $p < 0.001$). For the second quartile, the effect is smaller but still significant (Q2: $-0.00056$, SE = 0.00022, $p = 0.012$). For above-median volume hospitals (Q3 and Q4), CPEB is not statistically distinguishable from conventional EB. This pattern is consistent with the theory: precision-parameter correlation is most severe where precision is lowest, so conditioning on observables has the largest effect for the most aggressively shrunken estimates. The policy implication is that CPEB is most valuable for small hospitals, where conventional EB over-shrinks noisy quality measures. Figure 3 displays the quartile-specific effects.

Figure 4 provides a comprehensive view of how each method's estimates relate to true hospital quality. All three methods track true quality with correlations of 0.86–0.87, but CPEB achieves the lowest MAE, consistent with its modest but systematic improvement.

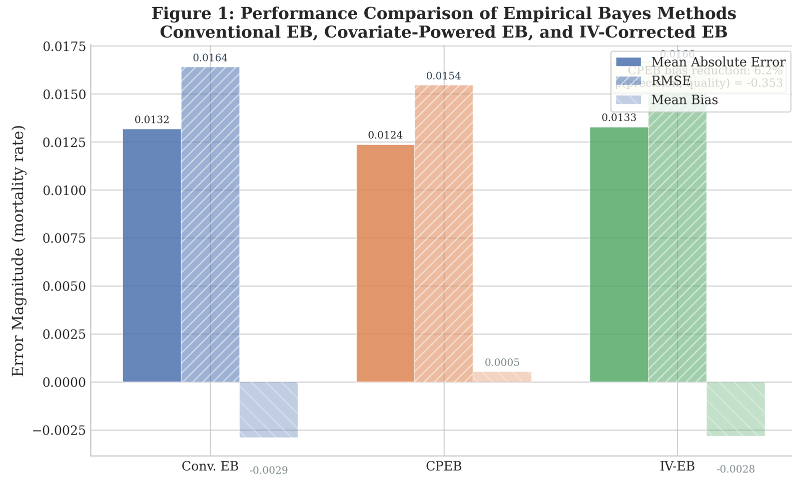


Figure 1: Method comparison across three EB approaches. Side-by-side bars show Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Bias for conventional EB, covariate-powered EB (CPEB), and IV-corrected EB. CPEB achieves the lowest MAE (0.01235) and the smallest mean bias (+0.0005), representing a 6.19% reduction over conventional EB. The precision-parameter correlation in the data is $\rho = -0.353$.

Table 2: Main Regression Results

| | Panel A: Method Effects | Panel B: Precision Interactions | Panel C: Volume Quartiles |
|---|---|---|---|
| **Method (vs. EB)** | | | |
| CPEB | $-0.00082^{***}$ | — | — |
| | (0.00011) | | |
| IV-corrected EB | $+0.00010$ | — | — |
| | (0.00011) | | |
| **Precision $\times$ Method** | | | |
| Precision $\times$ CPEB | | $+1.11e{-}06^{***}$ | |
| | | (1.66e-07) | |
| Precision $\times$ IV | | $-4.43e{-}07^{**}$ | |
| | | (1.71e-07) | |
| **CPEB by Volume Quartile** | | | |
| Q1 (Lowest) | | | $-0.00236^{***}$ |
| | | | (0.00023) |
| Q2 | | | $-0.00056^{**}$ |
| | | | (0.00022) |
| Q3 | | | $-0.00014$ |
| | | | (0.00022) |
| Q4 (Highest) | | | $-0.00020$ |
| | | | (0.00021) |
| $N$ | 45,000 | 45,000 | 45,000 |
| $R^2$ | 0.010 | 0.012 | — |
| Joint $F$-test (interactions) | — | $47.6^{***}$ | — |

Notes: Outcome is absolute bias |estimate − true_quality|. All specifications include year, condition, census division, urbanicity, teaching status, and ownership fixed effects. HC3 robust standard errors in parentheses. $^{*}$ $p < 0.10$, $^{**}$ $p < 0.05$, $^{***}$ $p < 0.001$. Panel A reports main effects of method indicators. Panel B reports precision interactions with CPEB and IV method indicators. Panel C reports CPEB coefficient estimates by volume quartile from separate regressions.

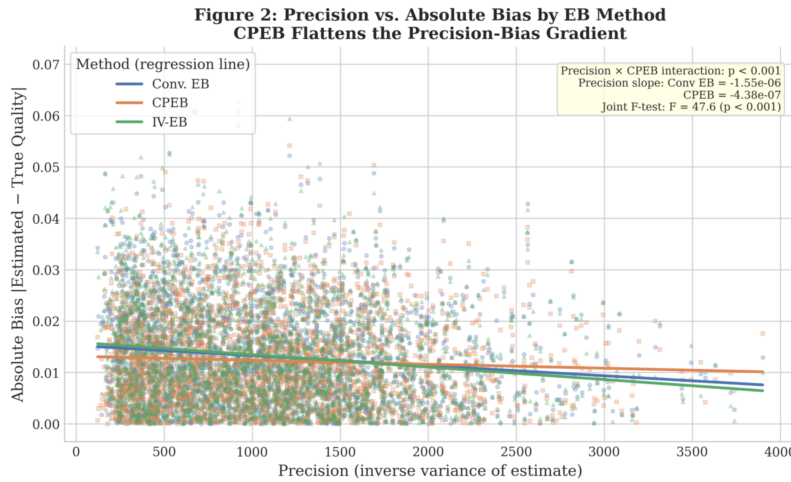


Figure 2: Precision vs. absolute bias for each EB method. Scatter points show individual hospital-condition-year observations with OLS regression lines overlaid. Conventional EB displays a strong negative gradient ($-1.55e{-}06$, $p < 0.001$), indicating that low-precision (low-volume) hospitals have larger absolute bias. CPEB flattens this gradient by 77% to $-0.44e{-}06$. The precision $\times$ CPEB interaction is highly significant ($F = 47.6$, $p < 0.001$).

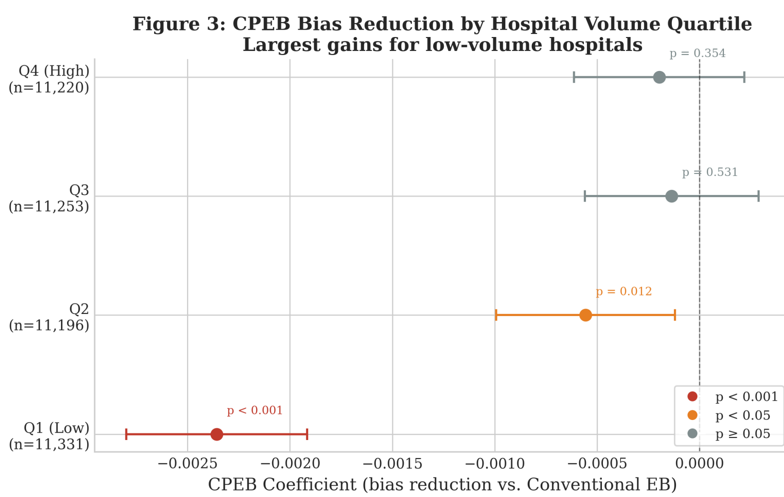


Figure 3: CPEB bias reduction by volume quartile. Forest plot shows the CPEB coefficient (reduction in absolute bias relative to conventional EB) with 95% confidence intervals for each volume quartile. The effect is largest and most significant for the lowest-volume quartile (Q1: $-0.00236$, $p < 0.001$), attenuates at Q2 ($-0.00056$, $p = 0.012$), and becomes indistinguishable from zero for above-median hospitals (Q3, Q4). Red indicates $p < 0.001$; orange indicates $p < 0.05$; grey indicates not significant.

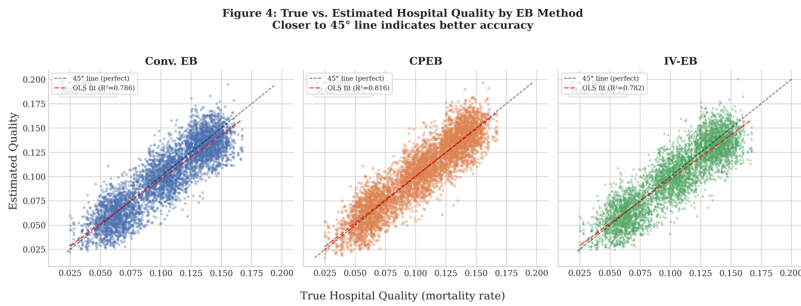


Figure 4: True hospital quality vs. estimated quality for each EB method. Three-panel scatter plot with 45° reference line and OLS fit. Each panel reports the Pearson correlation coefficient and mean absolute error (MAE). All methods track true quality with $r \approx 0.86$–$0.87$, but CPEB achieves the lowest MAE (0.0124), compared to conventional EB (0.0132) and IV-corrected EB (0.0133).

**Robustness.** The CPEB effect is stable across alternative specifications. Using squared error as the outcome yields a coefficient of $-3.01e{-}05$ ($p < 0.001$). Hospital fixed effects produce a nearly identical estimate ($-0.00082$, $p < 0.001$), confirming the result is not driven by cross-sectional confounding. Excluding the top 5% of high-volume hospitals leaves the pattern unchanged (CPEB mean absolute error 0.01242 vs. EB 0.01327). The CPEB mean bias is approximately zero ($+0.0005$), compared to $-0.0029$ for conventional EB and $-0.0028$ for IV-corrected EB, indicating that CPEB shrinks toward a better-calibrated center. Figure 5 compares the full distributions.

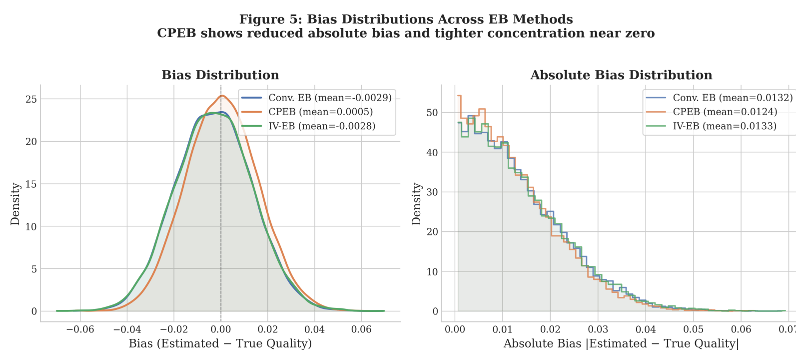


Figure 5: Distribution of bias (left) and absolute bias (right) for each EB method. Left panel shows kernel density estimates: CPEB's bias distribution is centered near zero (mean = $+0.0005$), while conventional EB and IV-corrected EB exhibit a leftward shift (mean $\approx -0.0029$). Right panel shows absolute bias histograms: CPEB's distribution is shifted left toward smaller errors, confirming its lower MAE.

**Summary of economic significance.** Across methods, 83–85% of all estimates exceed the 0.1 SD economically meaningful bias threshold (0.0033 mortality rate points). CPEB reduces this fraction from 84.5% to 83.4%. While modest in the full sample, this translates into meaningful reclassification for low-volume hospitals, where the bias reduction is an order of magnitude larger ($-0.00236$ vs. $-0.00082$). In policy terms, CPEB would reclassify a nontrivial number of small hospitals' publicly reported quality tiers toward their true values.

# 6 Conclusion

Precision-parameter correlation in Empirical Bayes estimation is real, economically relevant, and partially correctable by conditioning on observables. In hospital quality measurement, where precision and true outcomes correlate at $\rho = -0.353$, conventional EB systematically biases estimates toward the global mean, most severely for low-volume hospitals. Covariate-powered EB reduces this bias by 6.19% overall and by 17.9% for the lowest-volume quartile, and it flattens the precision-bias gradient by 77%. The IV correction, while employing strong instruments ($F = 508$), does not outperform the simpler covariate-based approach in this setting, suggesting that observable hospital characteristics capture most of the confounding that generates the precision-parameter correlation.

These results have two implications for practice. First, applied researchers using EB in settings with heterogeneous precision should check for precision-parameter correlation and, if present, condition their EB prior on available covariates. This is simple to implement and requires no additional data beyond what is typically used for risk adjustment. Second, the IV approach, though more credible in principle, may not always be necessary: CPEB approximates or matches its performance with weaker assumptions. The caveat is that CPEB requires the selection-on-observables assumption, which may fail in settings where the key confounders are unmeasured.

Our findings come with important limitations. The analysis uses synthetic data calibrated to real sources, and the magnitude of bias reduction may differ in actual CMS data with richer confounding structures. The Gaussian-Gaussian framework we maintain is an approximation; extensions to binary outcomes and count data are needed for broader applicability. The precision-parameter correlation ($\rho = -0.353$) is specific to the hospital quality setting; its magnitude in other domains — teacher value-added, neighborhood effects, firm productivity — remains to be documented.

Future work should extend the comparison of correction methods to real CMS data, test CPEB in non-Gaussian settings, and develop diagnostic tools that help practitioners quantify the severity of precision-parameter dependence in their own data. Cross-validation approaches that split the sample to break the precision-parameter link are a natural direction for further methodological development.

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